

Wahid Mahmood

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Research Interest

NLP, XAI, LLM, Agentic AI, Reinforcement Learning, Human AI Interactions.

Education

Khulna University of Engineering & Technology (KUET)

Jan 2019 – Feb 2024

B.Sc in Computer Science and Engineering (CSE)

- GPA: 3.53/4.0 ([Transcript](#))
- **Coursework:** NLP, Linear Algebra, Statistics, Data Mining, Image Processing and Computer Vision.

Publications

W. Mahmood & S. M. M. Ahsan, "A Semi-Supervised Approach For Detecting Agricultural Land Reduction In Bangladesh Using Satellite Images."

Sep 2024

IEEE International Conference on Computing, Communication, and Intelligent Systems (COMPAS), 2024. DOI: [10.1109/COMPAS60761.2024.10795941](https://doi.org/10.1109/COMPAS60761.2024.10795941)

- Developed a novel semi-supervised approach for agricultural land change detection, which significantly reduced the time and hardship required for manual annotation.
- Further evaluated the created dataset with other deep learning methodologies.

W. Mahmood, T. Joty & S. M. M. Ahsan, "NSNUNet: A Neighbourhood Attention Based Siamese Nested Unet for Change Detection using Satellite Images ."

Dec 2024

IEEE International Conference on Computer and Information Technology (ICCIT), 2024. DOI: [10.1109/ICCIT64611.2024.11022454](https://doi.org/10.1109/ICCIT64611.2024.11022454)

- Proposed a novel neighborhood attention-based UNet architecture, improving accuracy and efficiency in remote sensing change detection.

W. Mahmood, N. Hasan, M. Mahadi, S. Shuvo, T. Joty & N. Siddique, "Unveiling Overfitting Patterns: Explainable AI-Driven Benchmarking of LLMs, Deep Learning Models, and a Hybrid Framework for Medical Image Classification."

May 2026

In Preparation

- Formulated a methodology to assess the interpretability and decision-making rationale of both Deep Learning (DL) architectures and Vision-Language Models (VLMs).
- Designed an evaluation pipeline encompassing comprehensive image preprocessing, specialized dataset curation, and the development of a novel hybrid DL model to serve as a comparative baseline.

Thesis

Study on Agricultural Land Reduction in Bangladesh using Satellite Images

Mar 2024

- Collected and preprocessed raw remote sensing images over the area of Bangladesh from the Sentinel-2 satellite.
- Created agricultural land change detection dataset consisting of over 500 image pairs over Bangladesh region.
- Explored multiple ML techniques (XGBoost, SVM) and extended the open-source framework called Open-CD.

Experience

Machine Learning Engineer, ReliSource – Dhaka, BD

June 2024 – Current

- **Protocol Extractor:**
 1. Architected a high-precision, agentic Retrieval-Augmented Generation (RAG) pipeline to automate information extraction from complex clinical trial protocols.
 2. Integrated intelligent markdown header-based chunking, cross-encoder reranking, and other methodologies

to optimize accuracy up to 85% on BERTScore and semantic similarity metrics.

3. Included automated response validation and a Human-in-the-Loop (HITL) framework for gradual improvement using few-shot learning.
4. Full offline inference backend orchestrated via LangGraph, FastAPI, LM Studio, Qdrant.

- **Agentic Medical Inventory Manager:**

1. Built an end-to-end forecasting and restocking platform that pairs time-series forecasting with an agentic LLM decision loop.
2. Pulled news, weather, and holiday data to compute contextual demand multipliers and used integer Linear Programming to generate budget-constrained restocking recommendations.

- **AI Chatbot:** RAG-based customer service AI Chatbot using LangChain, FastAPI, and OpenAI, achieving a 4-second response time, including source citations and message streaming, by implementing efficient retrieval and answer-generation workflows.
- **Data Engineering Training:** Completed Azure Data Engineering training program and supported client data workflows by ingesting and processing data using Azure Data Factory (ADF) and Databricks.
- **Drug Discovery Pipeline:** ML-based drug discovery pipeline to screen and validate potential drug candidates, including ADMET analysis, utilizing tools like AlphaFold, DeepChem, Biopython, RDKit etc.

Personal Projects

Intent Based Product Search

Mar 2025

- Engineered an intent-aware product retrieval system for e-commerce sites utilizing Llama 3.2 1B (finetuned).
- Implemented a hybrid search architecture achieving sub-2000ms latency for e-commerce queries.
- Tools Used: Langchain, FastAPI, Langsmith, PGVector, Ollama, Docker, and Redis.

Open-CD

Feb 2024

- Modified version of the open-source change detection framework named Open-CD.
- Add the feature of loading images with more than 3 channels and running all the models on a specific dataset.
- Tools Used: Python, OpenMM, Rasterio.

DOKAN KOI?

July 2023

- E-commerce app connecting users with nearby stores, featuring shop location tracking using Firebase.
- Tools Used: Flutter, Firebase

Skills

Languages: Python, C/C++ , Java

AI/Machine Learning: TensorFlow, PyTorch, Scikit-learn, Transformer, OpenCV, statsmodels.

Frameworks and Tools: LangChain, LangGraph, CrewAI, FastAPI, React, Flutter, Langfuse, Langsmith.

Data & Vector Database: PostgreSQL, PGVector, Qdrant, Neo4j, Snowflake, Databricks

Others: Redis, Azure Data Factory, MLflow, Git/GitLab, LM studio, Ollama, Docker.

Standardized Test Scores

International English Language Testing System (IELTS): 7.5 (overall)

Aug 2025

Achievements

Deans Award (Offered to average GPA of 3.75 or above achievers)

2019 – 2020

- Awarded for academic excellence by achieving a GPA of 3.83.

7th at Intra KUET Programming Contest

2022

- Placed 7th out of 85 competing teams.

5th at Poridhi Presents AI Engineering Hackthon

2025

- Placed 5th out of 80 competing teams.

Online Competitive Programming

2019-2023

- **Codeforces:** Specialist (Max Rating: **1493**, 650+ problem solved.)
- **Codechef:** 4 star (Max rating **1864**).

References

Dr. Sk. Md. Masudul Ahsan

Academic Supervisor

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Khulna University of Engineering & Technology (KUET)

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